BeeQ: Response to Reviewers

Anonymous review copy

8 October 2026

We have revised the complete manuscript against both reviewer reports and the archived results. The revision preserves the original six-family X10 comparison and adds two explicitly retrospective class-weight sensitivity analyses for the MLP. It separates discrimination from toxic-compound recovery, clarifies the historical holdout's exposure, limits CR8 claims to individual cases, and explains the kernel geometry without equating high alignment with proven exponential concentration. Tables and figures have been rebuilt from the verified artifacts.

One experimental recommendation remains unfulfilled: expansion beyond X10. The revision states this limitation explicitly and makes no claim that new representation experiments were performed or that representation dimensionality has been causally isolated.

1 Reviewer 1

1.1 Toxic recovery deserves equal prominence

Reviewer concern (summary). The sensitivity drop outside Development should be as prominent as the quantum–classical comparison, particularly in the abstract and conclusion.

Response. We agree and now organize the paper around both questions. The abstract reports original historical-holdout sensitivity of 0.082–0.265 alongside the kernel comparison. The title, Introduction, Results III-C, and Conclusions emphasize transfer and toxic recovery. Table III provides discrimination, sensitivity, specificity, and confusion counts for all original models and the additional MLP variants. We distinguish the larger historical partition from the two-positive CR8 case series and avoid population claims based on CR8.

Location in the revised manuscript: Title; Abstract; Introduction; Results III-C and III-E; Conclusions; Table III; Figures 2, 3, and 5.

1.2 Connect geometry to theory and concrete adaptation methods

Reviewer concern (summary). Explain the relationship to kernel concentration and cite ways to adjust or train quantum kernels.

Response. Results III-D now connects the observations to concentration theory while stating what our experiment does not test. We use exact overlaps at a single qubit count; CKA measures finite-cohort alignment, not exponential scaling or a barren plateau. Off-diagonal variation remains measurable. The directly supported explanation is Product's near-Gaussian small-angle form and the RBF bandwidth control: effective ranks 3.086 and 3.078, and CKA 0.999794 between Product and that control. We also clarify that the predictive "matched RBF" does not have a forced equal bandwidth.

We discuss bandwidth selection, projected kernels based on reduced-state observables, and kernel-target alignment as concrete follow-up approaches. These were not evaluated here, and training an embedding is not claimed to guarantee mitigation. The cited primary sources are Shaydulin and Wild, *Phys. Rev. A* 106, 042407 (2022); Huang et al., *Nat. Commun.* 12, 2631 (2021); Hubregtsen et al., *Phys. Rev. A* 106, 042431 (2022); and Thanasilp et al., *Nat. Commun.* 15, 5200 (2024).

Location in the revised manuscript: Methods II-C and II-F; Results III-D; Limitations; Supplementary Tables D6a–D6c; Figure 4; references 3, 4, 27, and 28.

1.3 The original MLP balancing mismatch

Reviewer concern (summary). Repeat the MLP with balancing or clearly retain the mismatch as a limitation before interpreting its lower performance.

Response. We performed both a controlled rerun and a fully selected balanced analysis. The audit confirms that the original implementation omitted sample weights. Its recorded scikit-learn 1.9.0 supports weighted fitting, but repository history does not establish the authors' reason for omitting it. We therefore report an implementation omission with unresolved intent, not a proven library restriction or deliberate design decision.

The original predictions were first reproduced within 1.89×10^{-15}. Each balanced fit uses $w_c = N_{\text{fit}}/(2n_{c,\text{fit}})$ from its training labels only. No resampling or evaluation weighting is used. Analysis A fixes original selected hyperparameters; Analysis B repeats the original 12-candidate search with the same five outer/four inner cluster folds, preprocessing, model seeds, and selection criterion. Thresholds are independently estimated for each variant from training OOF scores using the original MCC and tie-breaking rule. Native stopping settings are preserved, including their interaction with weights and their molecule-level, non-grouped reservation.

Development mean AUROC is 0.677 originally, 0.688 in A, and 0.679 in B; AUPRC is 0.596, 0.609, and 0.594. Sensitivity does not improve consistently: it increases in four folds but drops in fold 3, and pooled MCC decreases. On the historical holdout, toxic recovery changes from 4/49 to 11/49 and 13/49, while false positives rise from 4 to 16 and 22. AUROC remains only 0.481–0.482 in the balanced variants. In CR8, both recover Isocycloseram, still miss Ethiprole, and add a Fluoxapiprolin false positive. Thus balancing affects threshold decisions but does not establish a consistent discrimination improvement or explain away the MLP's lower performance. The quantum–RBF comparisons are unchanged.

All 693 fits have recorded histories and zero captured training warnings; an independent complete rerun reproduces all tabular outputs exactly. The original unweighted row, original artifacts, balancing limitation, and retrospective status remain visible.

Location in the revised manuscript: Methods II-C–II-E; Results III-B, III-C, and III-E; Limitations; Table II; Table III; Figures 2, 3, and 5; Supplementary Tables M1–M7.

1.4 Historical holdout exposure limits inference

Reviewer concern (summary). Earlier inspection may have affected upstream choices, so the historical holdout should not be treated as a clean confirmatory test.

Response. We now state this limitation in the abstract, Methods, the holdout Results, and Limitations. Even with disjoint clusters and a final Development-only freeze, earlier exposure could have influenced curation or representation choices; the retained evidence cannot quantify that influence. We characterize the historical results as exploratory transfer evidence. CR8 has separately checked structural provenance, but only two toxic cases. We distinguish provenance from statistical precision rather than calling either set a definitive validation. The new balanced-model evaluations follow observation of both sets and are explicitly retrospective.

Location in the revised manuscript: Abstract; Methods II-B and II-E–II-F; Results III-C; Limitations; Conclusions; Figure 1.

2 Reviewer 2

2.1 Quantitative context from prior ApisTox results

Reviewer concern (summary). Provide a numerical comparison against published fingerprint, graph-kernel, GNN, and pretrained-model benchmarks.

Response. Table IV and Results III-F now provide that context from the primary ApisTox study, including both published split settings and the BeeQ RF result. They explicitly distinguish cohort, split, and uncertainty definitions. AUPRC is marked as unavailable where not reported by the source tables; it is not reconstructed. These numbers contextualize X10 without presenting a matched benchmark or attributing protocol differences to representation.

Location in the revised manuscript: Introduction; Results III-F; Table IV; reference 9.

2.2 Expanded representations and attribution of the X10 limit

Reviewer concern (summary). Test X15, X20, or fingerprints under the same pipeline to determine whether compact representation drives the observed limitations.

Response. We agree that a matched representation sensitivity experiment is needed to answer this question. It has not been performed in this revision, which preserves the fixed X10 experiment and addresses the identified MLP asymmetry. We cannot determine from the existing comparison whether an information bottleneck, the learning method, or their interaction is responsible for the transfer limitations. Methods II-A now makes clear that the ten-input budget was chemically informed and qubit-compatible, not chosen by predictive feature optimization. Limitations identifies the absent representation control, and the abstract and conclusion restrict the negative quantum finding to the tested X10 protocol. Richer inputs are proposed as an experiment, without claiming they will recover the missed cases.

This recommendation is therefore addressed through scope clarification and an explicit unresolved empirical limitation, *not* through a completed representation ablation. A subsequent study should preserve molecular memberships and nested selection while changing representation, reporting both discrimination and the toxic-recovery/false-positive tradeoff.

Location in the revised manuscript: Abstract; Introduction; Methods II-A; Limitations, first paragraph; Conclusions.

2.3 CR8 has only two toxic compounds

Reviewer concern (summary). The conclusion gives more rhetorical weight to the two CR8 false negatives than their sample size supports.

Response. We have reduced the CR8 interpretation to compound-level observations and removed the claim that these cases identify a general representation failure. Results III-E and Figure 5 also distinguish the original decisions from the retrospective balanced MLP decisions. The conclusion does not infer regional sensitivity, broad chemotype performance, or a missing toxicological mechanism from two positives. Domain membership is described as evidence of proximity, not a correctness guarantee.

Location in the revised manuscript: Abstract; Methods II-F; Results III-E; Limitations; Conclusions; Figure 5; Supplementary Table M5.

3 Additional directly related corrections

We corrected the abstract's historical AUROC range to include the original MLP (0.462–0.706), and the holdout assignment to RBF/Product/IQP-ZZ to 0.706/0.697/0.696. The original MLP's three-decimal AUROC SD is 0.124. Reconstructed Product–IQP-ZZ CKA is 0.999506, resolving the abstract/body discrepancy. We removed the contradictory "with and without entanglement" wording for the separable Product map, the duplicated hardware sentence, the duplicated Methods paragraph in Results, and unresolved Table Sx, bandwidth, and equation markers. Equations, references, tables, and figures now have explicit definitions and traceable source artifacts.

The remaining experimental limits are stated rather than silently removed: richer representations have not been tested; native MLP stopping is not cluster disjoint; new downstream evaluations are retrospective; CR8 is small; and no noisy or hardware quantum experiment has been performed.